

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B.Tech****Branch : Biomedical Engineering****Semester : VI****Subject Code & Name: BTBM603 Biomaterials and Artificial Organs****Max Marks: 60M****Date:23/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Biomaterial is a _____ a) Synthetic or natural c.) Only Natural b.) Only synthetic d.) Made from metal only	Remember	1
2	In general, which of the following is/are a natural polymer? a) Chitosan C) . Poly(vinyl alcohol b. Polyphosphazene d. Polyurethane	Remember	1
3	Which of the following cannot assist in identifying cells from different tissue origins? a) Cell morphology c) Cell colour b) Cell phenotype d) Cell behavior	Remember	1
4	_____ is one of the most important aspects of biomaterial-tissue interactions a.) Biocompatibility c.) Bioequivalence b) Bioavailability d) Bioluminescence	Remember	1
5	Which of the following bioceramics is NOT bioactive? a) Titania c) Coralline HA b) Synthetic HA d.Alumina	Remember	1
6	When a wound becomes infected with bacteria which cells are first needed to fight the infection? a) White blood cells c) Macrophages b) Red blood cells d) None of the above	Remember	1
7	Which device or implant is not used for internal fracture fixation? a) Plaster or resin cast c) Plates b) Wires d) Screws	Remember	1
8	Which category of tissue implant response is one where cells and surrounding tissues are killed (become necrotic)? a) Toxic c) Bioactive b) Nearly inert d) Biodegradable or bioresorbable	Remember	1
9	Which of the following device involve the flow of blood? a) Dialyser c) Intraocular lens b) Hip implant d) Dental Restorative	Remember	1

